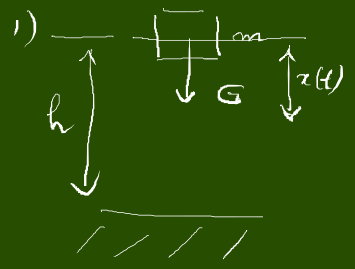
Ecuații diferențiale

Sisteme dinamice în timp continuu

1)



$ma = G = mg$

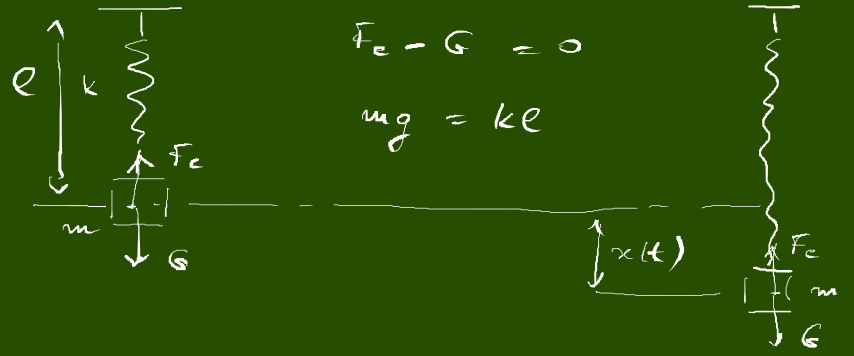
$a = g$

$x(t) \longrightarrow x'(t) = v(t) \longrightarrow x''(t) = a(t)$

deplasarea — viteza — accelerația

$x'' = g$

2)



$F_e - G = 0$

$mg = k\ell$

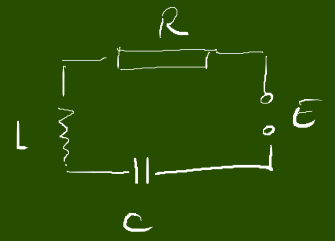
$ma = G - F_e$

$ma = mg - k(\ell + x(t))$

$ma = -kx(t)$

$mx'' + kx = 0$

3)



$E = U_R + U_L + U_C$

$U_R = R \cdot I$

$U_C = \frac{Q}{C} = \frac{\int_{t_0}^{t} I(s)\,ds}{C}$

$U_L = L \cdot I'(t)$

$I = I(t)$

$I(t) = \frac{Q(t) - Q(t_0)}{t - t_0} = \frac{\Delta Q}{\Delta t}$

$I(t) = Q'(t)$

$$\Rightarrow \quad L I'(t) + R \cdot I(t) + \frac{\int_{t_0}^{t} I(s)\,ds}{C} = E$$

$$L I''(t) + R I'(t) + \frac{I(t)}{C} = 0$$

Def. Ecuație diferențială de ordinul $\underline{I}$

$$x' = F(t, x)$$

$F : I \times D \longrightarrow \mathbb{R}$ cont.

$I \subseteq \mathbb{R}$ $\quad D \subseteq \mathbb{R}^n$

t variabila

$x = x(t)$ fct. necc.

Soluție a ec. dif. $x : J \subseteq I \longrightarrow D$ derivabilă

$$x'(t) = f(t, x(t)) \quad \forall\, t \in J$$

Problema Cauchy (Initial value problem) IVP

$$\begin{cases} x' = f(t, x) \\ x(t_0) = x_0 \end{cases} \qquad t_0 \in I,\ x_0 \in D$$

Thm. fundam. de existență și unicitate a soluției pb. Cauchy.

$$\begin{cases} x' = f(t, x) \\ x(t_0) = x_0 \end{cases} \qquad f : I \times D \longrightarrow \mathbb{R} \qquad I \subseteq \mathbb{R} \quad t_0 \in I \qquad D \subseteq \mathbb{R}^n \quad x_0 \in D$$

$f(t, x)$ și $\dfrac{\partial f}{\partial x}$ există și sunt / este cont.

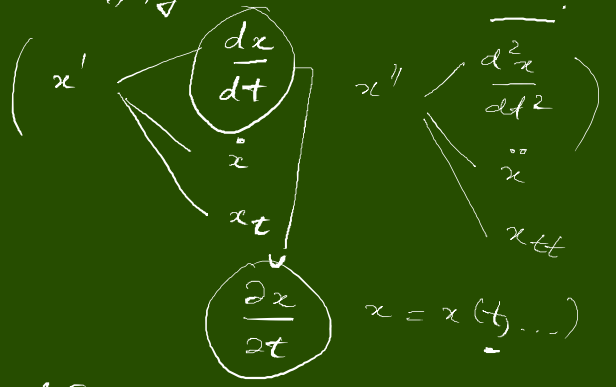
$\Rightarrow$ pb. Cauchy are sol. și sol. e unică în vecin. pct. (t_0, x_0)

$\exists!\ x = x(t) \qquad x : J \subseteq I \longrightarrow D \qquad t_0 \in J$

a. f. $\begin{cases} x \text{ deriv.} \\ x'(t) = F(t, x(t)) \quad \forall t \\ x(t_0) = x_0 \end{cases}$

Ecuação diferencial linear de ordinul I $\quad x' = f(t)x + g(t)$

$f, g : I \subseteq \mathbb{R} \longrightarrow \mathbb{R}$ cont.



I $\quad x' = \lambda x, \quad \lambda \in \mathbb{R} \qquad x = x(t)$

$\lambda = 1 \qquad x_1' = x_1 \qquad$ o soluție $x(t) = e^t$

$x_2' = x_2 \qquad\qquad x(t) = Ce^t$

$(x_1 + x_2)' = x_1 + x_2$

$\lambda \in \mathbb{R} \qquad \boxed{x' = \lambda x} \qquad x(t) = e^{\lambda t}$ o soluție

$x'(t) = \lambda x(t) \quad | : e^{\lambda t}$

$x'(t)e^{-\lambda t} - \lambda e^{-\lambda t} x(t) = 0$

$(x(t)e^{-\lambda t})' = 0 \quad \Rightarrow \boxed{x(t) = Ce^{\lambda t}} \quad C \in \mathbb{R}$

Exemplu. $x' = \lambda x - \dots$ creștera populației T. Malthus

$x(t) = Ce^{\lambda t}$ $\quad \lambda \simeq 1{,}05\%/\text{an}$

$t_0 = 2021$ $\quad x(t_0)$

$t_1 = ?$ $\quad x(t_1) = 2x(t_0)$

$$Ce^{\lambda t_1} = 2Ce^{\lambda t_0}$$

$$e^{\lambda(t_1 - t_0)} = 2 \qquad t_1 = t_0 + \frac{\ln 2}{\lambda} = t_0 + \frac{\ln 2}{0{,}0105}$$

66

Pb. Cauchy $\begin{cases} x' = \lambda x \\ x(t_0) = x_0 \end{cases}$ $\quad x(t) = Ce^{\lambda t}$ $\quad /\Rightarrow x_0 = Ce^{\lambda t_0} \Rightarrow C = x_0 e^{-\lambda t_0}$

$$\Rightarrow \boxed{x(t) = x_0 e^{\lambda(t - t_0)}}$$

II $\quad x' = f(t)x$ $\quad f : I \subseteq \mathbb{R} \longrightarrow \mathbb{R}$ continuă

Fie $F(t)$ primitivă a lui f, $F'(t) = f(t)$

$x(t) = e^{F(t)}$ o sol. a ec. dif.

$e^{F(t)} f(t) = f(t) e^{F(t)}$ ✓

$x'(t) = f(t) x(t) \quad | \cdot e^{-F(t)}$

$\underbrace{x'(t) e^{-F(t)} - f(t) e^{-F(t)} x(t)}_{\left(x(t) e^{-F(t)}\right)'} = 0$

$\left(x(t) e^{-F(t)}\right)' = 0 \Rightarrow x(t) = C e^{F(t)}, \; C \in \mathbb{R}.$

Pb Cauchy : $\begin{cases} x' = f(t) x \\ x(t_0) = x_0 \end{cases}$

F primitive pt f $\quad F(t) = \int f(t)\,dt$

$\begin{cases} f = F' \\ F(t_0) = 0 \end{cases} \longleftarrow F(t) = \int_{t_0}^{t} f(s)\,ds$

$\begin{cases} x(t) = C e^{\int_{t_0}^{t} f(s)ds} \\ x(t_0) = x_0 \end{cases} \Rightarrow x_0 = C \cdot e^{0} \Rightarrow C = x_0$

$\hookrightarrow \Rightarrow \boxed{x(t) = x_0 e^{\int_{t_0}^{t} f(s)ds}}$

III $\quad x' = f(t)x + g(t) \qquad f, g : I \subseteq \mathbb{R} \longrightarrow \mathbb{R}$ cont.

Metoda variației constantelor

Pas 1) $\quad x' = f(t)x \quad \Rightarrow \quad x(t) = Ce^{F(t)}, \; C \in \mathbb{R}$

↳ e. d. o. omog.

F primitiva pt f.

$F(t) = \int f(t)dt$.

Pas 2) $\quad x' = f(t)x + g(t)$ caut soluția de forma

$x(t) = C(t)e^{F(t)}$ unde $C(t)$ este

o fct. care urmează a fi determinată

$$C'(t)e^{F(t)} + C(t)e^{F(t)}f(t) = f(t)C(t)e^{F(t)} + g(t)$$

$$C'(t) = g(t)e^{-F(t)} \Rightarrow C(t) = \int g(t)e^{-F(t)}dt + k, \; k \in \mathbb{R}$$

$\Rightarrow$ sol. qcq. a ec. dif $x' = f(t)x + g(t)$ é

$$x(t) = e^{F(t)}\left(\int g(t)e^{-F(t)}dt + k\right) \qquad k \in \mathbb{R}.$$

Pb. Cauchy $\begin{cases} x' = f(t)x + g(t) \\ x(t_0) = x_0 \end{cases}$

$$F(t) = \int_{t_0}^{t} f(s)ds$$

$$x(t) = e^{\int_{t_0}^{t} f(s)ds}\left(\int_{t_0}^{t} g(\tau)e^{-\int_{t_0}^{\tau} f(s)ds}d\tau + k\right)$$

$$x(t_0) = x_0 \Rightarrow x_0 = e^{1}(0 + k)$$

$$\Rightarrow x(t) = e^{\int_{t_0}^{t} f(s)ds}\left(\int_{t_0}^{t} g(\tau)e^{-\int_{t_0}^{\tau} f(s)ds} + x_0\right)$$

Exemplo 1) $\begin{cases} (1+t^2)x' = 2tx + (1+t^2) \\ x(0) = -2 \end{cases}$

$$x' = \underbrace{\frac{2t}{1+t^2}}_{f(t)} x + \underbrace{1}_{g(t)}$$

Pas 1) $\quad x' = \underbrace{\frac{2t}{1+t^2}}_{f(t)} x$

$$x(t) = Ce^{F(t)} = Ce^{\ln(1+t^2)} = C(1+t^2)$$

$$F(t) = \int \frac{2t}{1+t^2}\,dt = \ln(1+t^2)$$

Pas 2) $\quad x' = \frac{2t}{1+t^2} x + 1$ $\qquad x(t) = C(t)(1+t^2)$

$$C'(t)(1+t^2) + C(t)\cdot 2t = \frac{2t}{1+t^2} C(t)(1+t^2) + 1$$

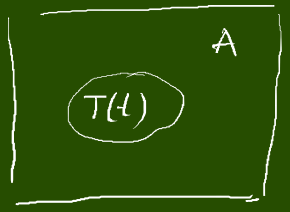
$$\Rightarrow C'(t) = \frac{1}{1+t^2} \Rightarrow C(t) = \int \frac{1}{1+t^2}\,dt = \operatorname{arctg} t + k$$

$\Rightarrow \quad x(t) = (1+t^2)(\arctg t + k) \quad \Big| \Rightarrow -2 = k$

$x(0) = -2$

$\Rightarrow \quad x(t) = (1+t^2)(\arctg t - 2)$

2)



$T' = -k(T - A)$ $\qquad k > 0$ $\qquad T(t) > A$

$T' = -kT + kA$

$T(t) = e^{-kt} \cdot C(t)$

$C'(t)e^{-kt} + C(t)e^{-kt}(-k) = -ke^{-kt}C(t) + kA$

$C'(t) = kAe^{kt}$

$C(t) = Ae^{kt} + B$

$\Rightarrow \quad T(t) = C(t)e^{-kt} = A + Be^{-kt} \xrightarrow{t \to \infty} A$

ex. $\quad A = 24^\circ C \qquad t_1 = 8{:}00\ AM \qquad T(t_1) = 28^\circ C \qquad t_0 = ?$

$t_2 = 9{:}00\ AM \qquad T(t_2) = 26^\circ C \qquad T(t_0) = 37^\circ C$

$$\begin{cases} 24 + Be^{-8k} = 28 \\ 24 + Be^{-9k} = 26 \end{cases} \Rightarrow \begin{matrix} Be^{-8k} = 4 \\ Be^{-9k} = 2 \end{matrix}$$

$$\overline{e^k = 2} \qquad k = \ln 2$$

$$B = 4e^{8k} = 4e^{8\ln 2} = 2^{10}$$

$$T(t) = 24 + 2^{10} \cdot e^{-\ln 2 \cdot t} = 24 + 2^{10-t}$$

$t_0 = ?$ $\quad T(t_0) = 37°C$

$$2^{10-t} = 13 \qquad 10 - t = \log_2 13 \simeq 3{,}7$$

$$t \simeq 6{,}3 \qquad \simeq 6{:}30 \text{ AM}$$

3) $x' = \lambda x - \mu x^2 \qquad \lambda, \mu > 0$

ecuația logistică

$$\lambda x - \mu x^2 = 0 \Leftarrow \begin{matrix} x(t) = 0 \\ x(t) = \frac{\lambda}{\mu} \end{matrix}$$

$x = x(t) \qquad y(t) = \frac{1}{x(t)} \qquad x(t) = \frac{1}{y(t)}$

$$-\frac{y'}{y^2} = \lambda \cdot \frac{1}{y} - \mu \cdot \frac{1}{y^2} \quad | \cdot (-y^2)$$

$$y' = -\lambda y + \mu .$$

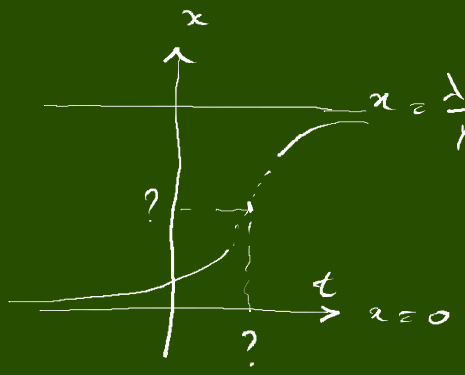
$y(t) = c(t) e^{-\lambda t}$ $\quad c'(t)e^{-\lambda t} - \lambda e^{-\lambda t} c(t) = -\lambda c(t) e^{-\lambda t} + \mu$

$\Rightarrow c'(t) = \mu e^{\lambda t}$

$c(t) = \dfrac{\mu e^{\lambda t} + k}{\lambda}$

$\Rightarrow y(t) = c(t) e^{-\lambda t} = \dfrac{\mu + k e^{-\lambda t}}{\lambda}$

$\Rightarrow x(t) = \dfrac{\lambda}{\mu + k e^{-\lambda t}}$



fct. sigmoïdale / sigmoïda

$\lim_{t \to -\infty} x(t) = 0$

$\lim_{t \to \infty} x(t) = \dfrac{\lambda}{\mu}$

$\lambda = \mu = 1$ $\quad k = 1$ $\quad x(t) = \dfrac{1}{1 + e^{-t}} = \dfrac{e^t}{1 + e^t} = \dfrac{1}{2} + \dfrac{1}{2} \tanh\left(\dfrac{t}{2}\right)$

$= \dfrac{\mathrm{sh}\left(\frac{t}{2}\right)}{\mathrm{ch}\left(\frac{t}{2}\right)}$

$e^{it} = \cos t + i \sin t$

$\cos t = \mathrm{ch}(it) = \dfrac{e^{it} + e^{-it}}{2}$